

# Read It Now, Not Later! Integrating Semantic Recommendation into In-Browser Web Page Scrap Tool

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## ABSTRACT

“Read-It-Later” is a generic term for web page scrap tools. However, research in Personal Information Management (PIM) has mainly focused on bookmarks, and little is known about Read-It-Later, especially when users interact with various scrap tools and devices. This study aims to understand users’ mental model of Read-It-Later and derive design implications that account for a new in-browser web page scrap tool, called “Read-It-Now.” Specifically, we conducted semi-structured interviews to develop the mental model of web scraping behaviors and extracted three implications (tool hierarchy, contextual aids, and proactive triggers) for encouraging the creation and retrieval of scraps. Based on the implications, we designed “Read-It-Now” which focuses on semantic recommendation: it detects the title of the currently activated tab and searches for the semantically closest scraps. We checked the empirical potentiality of the suggested prototype by testing with a sample dataset, and this provided insights for practitioners to design Read-It-Later tools.

## CCS CONCEPTS

• **Human-centered computing**; • **User studies**; • **Interface design prototyping**; • **Web-based interaction**;

## KEYWORDS

personal information management, information scrap, user interface, prototyping, semantic recommendation

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## 1 INTRODUCTION

Personal Information Management (PIM) strategies vary from person to person, but the majority of information is now digitally

stored in the internet era. Information scrap is a term that represents individual pieces of information that exist outside of a person’s primary management tool [1]. This study focuses on URL-based information scraps, which are equivalent to websites or web pages. Bookmark, a “file-system like” functionality provided in NETSCAPE-bookmark-file-1 format [2], is possibly the oldest and most conventional digital PIM tool for web page scrap since the introduction of the first World Wide Web [3]. PIM tools like Reading List (Chrome), Favorites (Safari), and Collection (Microsoft Edge) have been added to internet browsers as technology has grown as users’ information-gathering needs have become more complex. People also use various non-browser tools, such as pocket [4], messenger, memo, and note applications. These scrap tools for saving web pages, including bookmarks, are collectively referred to by the generic term “Read-It-Later.”

What actually happens when people save information to Read-It-Later? Several studies have been conducted to answer this question, and it was revealed that people continually accumulate new scraps and struggle with retrieving existing information because they frequently leave scraps disorganized and are overconfident in their memory [5-8]. However, these PIM studies have been limited to merely bookmarks [5-8], neglecting other types of Read-It-Later tools in various devices. This research gap shows the requirement to investigate how users interact with Read-It-Later tools. This investigation could enable us to understand users’ usage patterns and provide practical design insights to facilitate revisitation.

This study aims to suggest a viable solution to solve a retrieval issue, “Read-It-Now”, based on the investigation of users’ web page scrap tool in the form of mental model. To sum up, we first investigate users’ mental model for complex web page scraping behaviors through a semi-structured interview and derive design implications. After then, we establish primary design goals based on the findings in the study for designing our proposed interface. Finally, we develop “Read-It-Now,” an in-browser scrap tool to increase the discoverability of scraps and to facilitate user retrieval with recommendations based on semantic similarity.

## 2 USER RESEARCH: SEMI-STRUCTURED INTERVIEW

We developed research questions to get realistic insights in how to make people use Read-It-Later tools more actively.

(RQ1) Do individuals have different mental models when using different Read-It-Later tools?

(RQ2) How could users be encouraged to create more scraps?

(RQ3) How could users be encouraged to retrieve more scraps?

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## 2.1 Method

**2.1.1 Participants.** Ten interviewees (Age: mean=28.3, SD=2.54; Occupation: Students and Office Workers) were participated in the semi-structured interview to investigate the cross-device and cross-tool usage patterns for web scrap tools for approximately 90 minutes. Two of the participants were interviewed in person, and the other ten participants were interviewed remotely. They were priorly asked to bring their own electronic devices (e.g., PC, tablet PC, and smartphone). All the interviewees agreed to capture and record their shared screens, excluding their faces. The interviewees were compensated 15,000 KRW for their participation.

**2.1.2 Interview Design and Procedure.** After a brief instruction with consent for their participation, the pre-interview session was begun with asking general questions on the usage of the internet, browsers, devices, and scrap tools. Next, at the first phase, participants were asked to arrange virtual post-its according to eight questions for prioritizing their scrap tools used in each device (see Table 2 in Appendix A.1). For this work, Google Slides were used to display the real-time classification of virtual post-its. Each post-it label identified a scrap tool, and the questions inquired about the frequency of scrap creation and visitation of each tool on desktop, tablet, and mobile devices. Participants were asked to think aloud their thoughts during prioritization and explain reasons for their decisions.

At the second phase, a semi-structured interview was conducted following the pre-organized list of questions and accompanying open-ended discussion. Questions were developed to reflect the fundamental life cycle of information scrap [1, 9]: *Capture, Translate, Organization, Reuse and Delete*. According to [1], the information scrap is captured in a physical or digital form, translated to another form that is suitable for its purpose, organized with systematic classification or additional data for later use, and reused by the user. We added the stage of delete for our interviews because CRUD (Create, Read, Update and Delete) is known as a fundamental life cycle of online contents [10].

## 2.2 Results

**2.2.1 Device and Scrap Tool Usage. Internet Usage.** Participants used computers daily for an average of 7.8 hours (SD=3.12) and used the internet daily for an average of 6.4 hours (SD=2.50). They used the internet for work or study with a mean ratio of 75% (SD=0.16). **Device Ecosystem.** Five people had the same ecosystem among their devices (PC, tablet, and smartphone), while the other half had inconsistent ecosystems, including iOS and Windows/Android. **Browser Usage.** There were 10 Chrome users, 5 Naver Mobile users, 5 Safari users, 4 Naver Whale users, 1 Samsung Internet user, and 1 Microsoft Edge user. Participants utilized a mean of 2.75 internet browsers (SD=1.05). **Scrap Tool Usage.** Participants utilized 7.6 scrap tools in average (SD=1.42). All ten individuals utilized “Kakao Talk Chat with Myself” and “Notion” together. Appendix A.2 (Table 3) depicts the number of participants using each scrap tool. For reference, “Kakao Talk” is a dominant messenger application in South Korea. Interestingly, tool-free scrapping techniques were also mentioned during the interview.

**2.2.2 Users’ Mental Model.** A mental model of web scraping behaviors was established based on Indi Young’s structure of Mental Space-Task Tower-Task-Atomic Task-Supporting Quotes [11]. First, we divided transcripts into a total of 594 speech chunks and extracted minimal user tasks from them. After then, each task was categorized by related tool names and life stages. Next, tasks with similar patterns were grouped by a higher level of task tower. Finally, mental spaces, an abstract concept of fundamental user purpose, are constructed through the bottom-up process. After analyzing the transcripts, we rebuilt the life cycle of scrap in smaller stages: encounter, create, revisit, reuse and organize. ‘Translate’ and ‘Delete’ were incorporated into the organizing stage because, in the interview, users regarded them by means of organization. We drew key findings from each mental space to discuss about research questions. The mental model is summarized in Table 1 and detailed version is described in Appendix A.3 (Table 4).

**2.2.3 Discussion and Design Implications. RQ1: Do individuals have different mental models for different Read-It-Later tools?** Related key findings for this section are (a), (b), (e), (i), (j), (l) in Table 1. The interview found that when users decide which tool to use to create a scrap, a mental hierarchy is constructed. First, where the majority of scraps are initially stacked is uttered as “a temporary warehouse” or “add to cart” by participants. As these terms imply, they are primarily used to store links for short-term use temporarily. We would refer this initial place of a scrap as “primary storage.” Tools like screenshot, messenger, and memo applications belong here because they are highly accessible, portable, and speedy on any occasion. Over a period of time, the “stacked up” scraps are re-classified by the user and transferred to another scrap tool that is suitable for the character of each scrap and convenient to reuse. We would call this arriving place of the scrap as “secondary storage.” Secondary storages are utilized for archiving and long-term use. They have two distinct roles and purposes: shortcut and archive. Tools like bookmarks and favorites are perceived as “shortcuts” where frequently accessed scraps are stored. “Archive” is a place where users arrange related scraps in a single page and add contextual/visual information to summarize and beautify. Users also add their own systemic organization (e.g., Folder, Toggle, Contextual Information). The dominant example of a tool as an archive is Notion. Both work- and personal-related contents are mixed in primary storages and secondary “shortcuts,” but a secondary “archive” is usually a place for work-related scraps because they require time to study, summarize and organize the content inside. In this way, scrap tools could be classified in the mental hierarchy of users. Different characteristics of tools lead to differences in their usage and mental model. Thus, the design implication is derived as follows: (D1) *The design strategy of the tool must be differentiated based on its mental hierarchy.*

**RQ2: How could users be encouraged to create more scraps?** Related key findings for this section are (c), (d), (j), (k) in Table 1. During the scrap creation stage, decisions are made in a minimal amount of time, considering many contextual aspects. When deciding where to store a scrap, a user should consider the context of the web page, the context of the tool, and the external context of future revisit scenarios altogether. According to the channel factor [1], minor activities, such as turning on the computer or

**Table 1: Mental spaces, task towers, tasks, and key findings of web page scrap**

|                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Mental Space: Encounter</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Task Tower- Tasks              | <ol style="list-style-type: none"> <li>1. Personal recommendation - Conversation/Lectures/Online chatting</li> <li>2. Web surfing - Shopping/Leisure</li> <li>3. SNS Following - Online community/Contents subscription</li> <li>4. Work(Study) - Research for work/Search for questions</li> </ol>                                                                                                                                                                   |
| Key Findings                   | <ol style="list-style-type: none"> <li>(a) Work-related contents were regarded as more time-consuming, archival, and suitable for PC.</li> <li>(b) Personally intriguing contents were regarded as more temporal, less important, and suitable for portable device.</li> </ol>                                                                                                                                                                                        |
| <b>Mental Space: Create</b>    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Task Tower- Tasks              | <ol style="list-style-type: none"> <li>1. Determine whether to scrap the web page or not - Read the page/Perceived usefulness/Importance</li> <li>2. Determine where to keep - Available time/Ease of use/Need for sharing/Page context</li> </ol>                                                                                                                                                                                                                    |
| Key Findings                   | <ol style="list-style-type: none"> <li>(c) Decisions are made in a very limited of time while taking many contextual features into account.</li> <li>(d) Users don't want to break the flow of work when creating a scrap.</li> <li>(e) Majority of scraps are first saved in scrap tools with high accessibility and reexamined later.</li> </ol>                                                                                                                    |
| <b>Mental Space: Revisit</b>   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Task Tower- Tasks              | <ol style="list-style-type: none"> <li>1. Intention to find a scrap arises - New task starts/In communication/Need for work</li> <li>2. Search for the desired scrap - Keyword search/Locational navigation/Thematic hierarchy/Date</li> <li>3. Explore a tool aimlessly - Daily Routine/On the road/When get bored/Working/Lost in scraps</li> </ol>                                                                                                                 |
| Key Findings                   | <ol style="list-style-type: none"> <li>(f) Most retrieval triggers occur during external communication with people, arising the need for shareability.</li> <li>(g) Internal triggers are highly dependent on user's memory.</li> <li>(h) Long list of scraps causes cognitive load of users makes it hard to remember, thus leading locational search.</li> </ol>                                                                                                    |
| <b>Mental Space: Reuse</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Task Tower- Tasks              | <ol style="list-style-type: none"> <li>1. Summarize/Organize - Additional Memo/Beautify arrangements/Archive URL</li> <li>2. Share the summarization - Release for all/Share directly with specified people</li> <li>3. Share the link - Messenger Application/SNS/To his(her) another device</li> </ol>                                                                                                                                                              |
| Key Findings                   | <ol style="list-style-type: none"> <li>(i) When the scrap is for archival use, users want to arrange relative scraps as a page, add information and beautify.</li> <li>(j) Users often want to be prepared in case of they need to share links for others.</li> <li>(k) Transmission among devices or tools makes compatibility and shareability important.</li> </ol>                                                                                                |
| <b>Mental Space: Organize</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Task Tower- Tasks              | <ol style="list-style-type: none"> <li>1. Transfer to an adequate tool - Move to secondary tools/After moving, discard the scrap from temporary tool</li> <li>2. Organize for findability - At the moment of creation/Sort by thematic categorization/Change folder structure</li> <li>3. Get affective influence - Proud, satisfied/Indebtedness, shame, burden</li> <li>4. Delete a scrap - Discard outdated scraps/Leave them unless they irritate eyes</li> </ol> |
| Key Findings                   | <ol style="list-style-type: none"> <li>(l) Reexamined scrap gets moved to a suitable tool according to its role and purpose: shortcut or archive.</li> <li>(m) Scraps are sometimes hard to be classified into a single folder or tag and causes difficulties to users.</li> <li>(n) User often forgets about scraps in visual blind spots(e.g., bottom of the list, folder in folder).</li> </ol>                                                                    |

trying to find a scrap application on the phone, can yet operate as strong facilitators or impediments. Therefore, scrap tools should have context-supportive features that reduce psychological burdens and function as a facilitator of creating a scrap. In the study of Jones et al. [5], highly valued functions in the decision of which scrap tool to use are the number of access points, portability, context, and reminding. In the interview, we found that shareability also acts as a crucial concern. As explained in RQ1, users often transfer scraps across tools and device. The mental model also shows that the stage of reuse often comes with a need to share. Accordingly, a design implication is derived as follows: *(D2) The functionality of scrap tools should be advantageous to the user's context-specific issues.*

**RQ3: How could users be encouraged to retrieve more scraps?** Related key findings for this section are (f), (g), (h), (m),

(n) in Table 1. Two problems in retrieval were found in the analysis of the mental model. First, in the revisit stage, external retrieval triggers mainly occur in real life, often during communication with other people. These triggers are not controllable and dependent on coincidences. Internal triggers (i.e., the need for study) were also highly dependent on the user's memory. It implies that if a user forgets about a scrap, the page becomes outdated and is never revisited. The second problem is the difficulty in scrap navigation. If user remembers precise keywords of the scrap title, they use direct keyword search. Otherwise, they mostly used their own thematic organization system (i.e., folder) or just navigated one by one. We found that a single scrap often cannot be classified into a single folder in most cases because it has multiple attributes in the mind of the user, such as "recipe," "to-do," and "christmas." It made users distressed and causes the organization of scraps to be prolonged.

An unorganized list of web page scraps also makes scraps from being accessible because it creates visual blind spots (e.g., bottom of the list, folders in folders). In summary, scrap tools need to provide proactive triggers to remind users of scraps in purpose of increasing the recall and searchability. The design implication is derived as follows: (D3) *Proactive reminder triggers should be provided to increase retrieval and ease of search.* (D4) *Multiple organizational strategies are needed for ease of search.*

## 2.3 Design recommendations

**2.3.1 (D1) The design strategy of the tool must be differentiated based on its mental hierarchy.** Whether the scrap tool is positioned in primary or secondary storage determines which design strategy should be prioritized. Characteristics such as accessibility, portability, and ease of editing determine its role. Desirable advantageous functions in primary and secondary storage as shortcuts are *portability, number of access points, compatibility, speed, and shareability with myself*. Desirable advantageous functions in secondary storages as archives are *rich contextualization, organization, visualization, representation, and shareability with others*.

**2.3.2 (D2) The functionality of scrap tools should be advantageous to the user's context-specific issues.** Participants mentioned examples of supportive functions that add contextual conveniences in the process of scrap lifecycle. Firstly, *a simple and speedy creation interface* without page movement is essential to allow users to maintain the flow of work as much as possible. A lightweight organization function at the moment of creation is also efficient. Second, *automatic preview loading* was a feature that participants loved. When an URL is added to the list, the tool automatically crawls the thumbnail, title, and a piece of the body paragraph and visualizes them as a preview. Participants especially preferred large thumbnails and body paragraph previews because they help users not to leave a note about the content and recall memory. Lastly, *a convenient transfer* feature was needed for increased shareability. In the case of primary storage, sharing a single link is enough, but secondary functions need a feature to export multiple links at once.

**2.3.3 (D3) Proactive reminder triggers should be provided to increase retrieval and ease of search.** We have demonstrated the lack of proactive triggers. This problem is originated from the storage-centric characteristics of scrap tools: if the place where scraps are stored is separated from where scraps are used, it causes users to move back and forth across windows and users are also required to remember where they stored a scrap and how to find it [12–14]. Examples of task-centric design strategies in [14] are giving visual salience to the currently activated task or context-aware automatic organizing suggestions. As such, *more task-centric triggers* can possibly lower users' retrieval costs.

**2.3.4 (D4) Multiple organizational strategies are needed for ease of search.** A combination of multiple organizational strategies alleviates organizational burden and enables easy navigation. Keyword search and foldering are the most popular organizational ways, but they are not enough for scrap tools as explained in 2.2.3. Instead, scrap tools should *combine those traditional methods with location-based search methods* like multiple hierarchies and visual filtering.

## 3 READ-IT-NOW: INTEGRATING SEMANTIC RECOMMENDATION INTO IN-BROWSER SCRAPS

### 3.1 Overview

In this section, we propose the early prototype of our in-browser scrap tool, “Read-It-Now”, based on the design implications found in the user research. Read-It-Now mainly aims to facilitate users create and revisit as many scraps as possible. This purpose corresponds to the role of a “primary storage.” Specifically, we tried to achieve the following specific design goals throughout the lifetime of web page scraps:

[G1] Set design strategies adequate for the primary storage. (Recommended from D1)

[G2] Allow an effective task-centric reminder to trigger retrieval. (Recommended from D3)

[G3] Allow for simple creation, automatic visualization, and convenient transfer. (Recommended from D2)

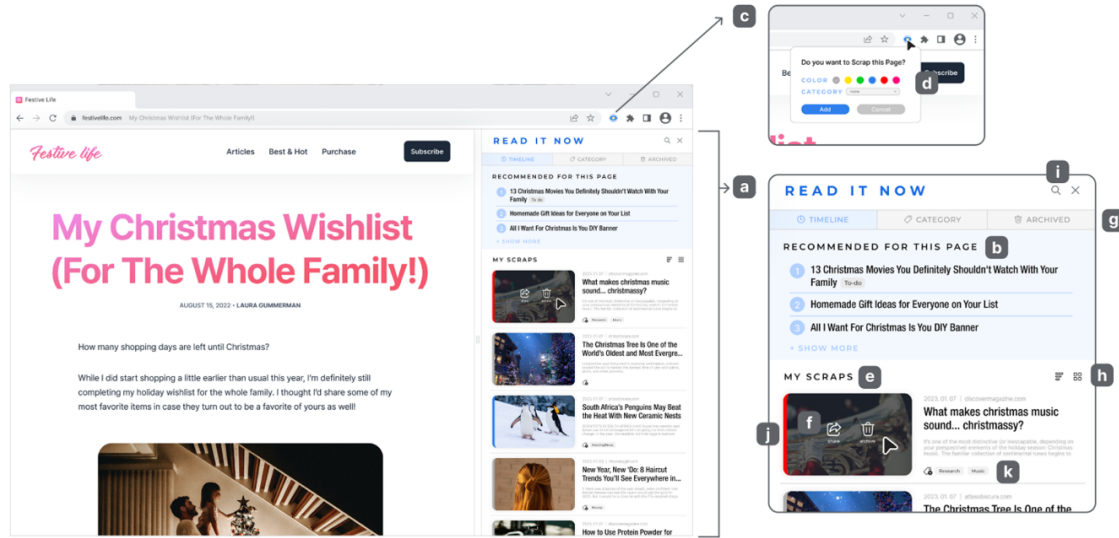
[G4] Allow multiple navigation strategies for ease of search. (Recommended from D4)

We also designed functions in accordance with our design goals. Functions in the interface are briefly described in Figure 1 and discussed in detail in the following section.

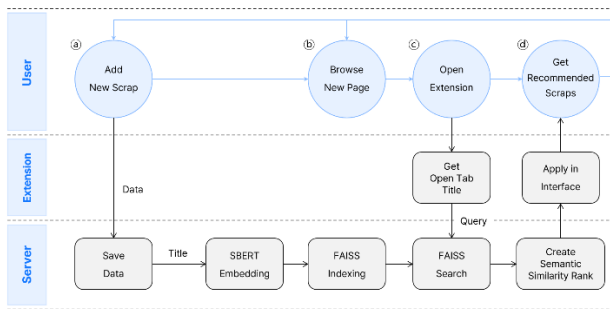
### 3.2 Interface Design and Development

**3.2.1 [G1] Design strategy for a primary storage.** G1 is the fundamental design goal of Read-It-Now and overall design decisions were made accordingly. Since primary storages value speed than quality contextualization, we designed three primary functions as follows: storing web page scraps, supporting scrap classification, and presenting semantic scrap recommendations. Functions to beautify or additive editing are not included in Read-It-Now. Providing a proactive task-centric trigger by adopting the concept of semantic recommendation is the most valued feature, and other interface elements were also designed to be consistent with users' mental model.

**3.2.2 [G2] Semantic recommendation as a task-centric retrieval trigger.** Read-It-Now is designed as a web extension, an in-browser application that can simultaneously interact with web browsers via an extension APIs, to detect the user's open tab and provides retrieval suggestions. The extension works in the browser side panel (Figure 1-a), which is located in the same window of the user's workplace, preventing users from moving back and forth to find scraps. Furthermore, semantic recommendation helps users remind scraps without cognitive effort. These features make Read-It-Now a task-centric scrap tool that overcomes limitations of a storage-centric tool. To create a conceptual prototype, we first gathered a collection of sample scraps. To represent a potential range of users' interests, 200 websites were randomly selected from 10 media blogs [15] in the categories of Hobbies, Style, New, Technology, and Trivia. Detailed data source is in Appendix A.4 (Table 5). The sample scrap file was transformed to a .CSV file containing two columns: title and url. To generate a semantic recommender system, it is necessary to preprocess and embed all scrap titles into semantic vectors so that each sentence can be positioned in the vector space. The distance between two sentences in the semantic vector space indicates their



**Figure 1: Overview of the Read-It-Now interface design.** Functions for G2 are usage of side panel in browser (a) and integrating semantic scrap recommendation as a task-centric retrieval trigger (b). Functions for G3 are simple and speedy creation (c, d), automatic preview loading (e), and convenient transfer (f). Functions for G4 are multiple navigation strategies (g, h, i, j, k).



**Figure 2: Workflow of the Read-It-Now**

similarity. We used lowercase, stopwords, and punctuation removal for preprocessing. We also used SBERT(Sentence-BERT), a state-of-the-art pre-trained transformer network for semantic sentence embedding [16]. After embedding, the FAISS library of Facebook AI Research was used to search for semantically nearest neighbors [17]. The similarity searches in FAISS utilized the IndexFlatIP index type and default Euclidean distance.

Figure 2 shows the workflow of the system. When a user adds a new web page (Figure 2-Ⓐ) to Read-It-Now, the title of the saved scrap is converted to a semantic sentence vector by SBERT and indexed by FAISS for future search convenience. User continues to browse the internet (Figure 2-Ⓑ) and comes across an intriguing page about which he or she wants to learn more. The user can then open the Read-It-Now extension (Figure 2-Ⓒ) and allow it to detect the browser’s open tab title. The title acts as a query for FAISS semantic search and returns similarity rankings for pertinent scraps (Figure 2-Ⓓ).

**Prototype test.** We put “My Christmas Wishlist (For the Whole Family!)” as a title query, and the search took 0.0135 seconds. Top-rated scrap titles were as follows: “13 Christmas Movies You Should Never Watch With Your Family” won the first place. “Homemade Gift Ideas for Everyone on Your List” came in the second place. “All I Want for Christmas Is You DIY Banner.” Came in the third place. Another test query was “Your 2023 Horoscope Is Here, And It’s A Year For Personal Power.” The search required 0.0096 seconds. The winning entry was titled “10 of the Largest Video Game Adaptations Coming in 2023.” “New Year, New ’Do: 8 Haircut Trends You’ll See Everywhere in 2023” came in the second place. “MY 2022 HOLIDAY GIFT GUIDE FOR TRAVELERS” placed third.

We created the example of interface design using the prototype test results (see Figure 1). The highlighted “Recommended for this page” section (Figure 1-b) at the top of the interface evaluates semantic similarities of existing scrap titles with the currently displayed page title and present recommendations.

**3.2.3 [G3] Simple creation, automatic visualization, and convenient transfer.** Read-It-Now provides quick access via a favicon in the browser’s toolbar. The app icon is a one-click button to add the page as a new scrap entry (Figure 1-c). In the pop-up window, users can select the scrap’s color scheme and category with relative ease (Figure 1-d). The process can be bypassed because the default settings are in place. The “My scraps” list is updated with a new entry (Figure 1-e). The scrap list is basically displayed in chronological order. When a new scrap is added to the list, Read-It-Now automatically crawls the HTML data of the page to obtain the image thumbnail, title, URL, and beginning of the paragraph to generate a preview. Because people preferred thumbnails as a reminder hint the most, images are arranged to take up the largest space. The “Share” and “Archive” buttons appear when users hover over the thumbnail(Figure 1-f). The URL link of the scrap is automatically

copied to the clipboard when the “Share” button is clicked, allowing for convenient transfer. The “Archive” function is explained in the next section.

**3.2.4 [G4] Multiple navigation strategies.** The tab bar (Figure 1-g) displays three distinct scrap classifications. Initially, “Timeline” merely stacks scraps so that users can easily scroll down the list by date and find what they seek. This location-based navigation is a prevalent design strategy for the preliminary storages. Read-It-Now enables several additional navigation strategies for extra convenience. People can apply different sort rules (Oldest, Latest, Colors) and view even with larger thumbnails (Gallery View) for those who prefer image previews than text (Figure 1-h). Keyword search is also available (Figure 1-i). Each scrap may have a single-color schema (Figure 1-j) and numerous categories to reflect multiple thematic hierarchies (Figure 1-k). The “Category” menu of the tab bar allows users to filter by categories. Lastly, “Archive” menu is where discarded scraps go. Users occasionally hoard scraps that they might see again “just in case” and are reluctant to delete them. Read-It-Now enables archiving function to relieve users of the burden of deletion, thereby reducing visual blind spots and stress.

## 4 CONCLUSION

This study investigated the comprehensive user behavior of cross-device and cross-tool interaction of Read-It-Later tools. Based on the findings of this investigation, we developed a user-centered in-browser scrap tool, Read-It-Now. Its theoretical contribution lies on filling the research gap of bookmark-centered research model in Personal Information Management. The user-centered mental model shed light on design implications for reducing users’ cognitive load and facilitating scrap retrieval. This study is expected to give insights for practitioners to design information scrap tools. However, the small sample size and lack of user personas for reading strategy remain limitations. Read-It-Now is still a prototype that is currently undergoing development. Our proposed interface in this study will be extensively developed and evaluated with the fundamental concept of semantic recommendation and a mobile interface to explore insights regarding cross-device interaction would be developed. Plus, our future work will be an empirical study that evaluates the efficiency and usability of the Read-It-Now by assessing the click rate, retrieval rate and read time of scraps compared to a scrap tool without semantic recommendations.

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## A APPENDICES

### A.1 Interview Instructions in the phase 1 of Semi-structured Interview

**Table 2: Detailed interview instructions in phase 1**

| No. | Instruction                                                                                                                      |
|-----|----------------------------------------------------------------------------------------------------------------------------------|
| 1   | Please remove the web page scraping tool that you use from the suggested post-it note and add any missing tools.                 |
| 2   | Among the responses in 1, please only leave the PC-related tools. Please indicate which tools you prefer the most.               |
| 3   | Among the responses in 1, please only include the mobile/tablet-specific tools. Please indicate which tools you prefer the most. |
| 4   | Please arrange the post-it notes according to the frequency with which scraps are generated on your computer.                    |
| 5   | Please arrange the post-it notes in the order of frequency of use on your computer.                                              |
| 6   | Please arrange the post-it notes according to the frequency with which scraps are generated on your mobile device or tablet.     |
| 7   | Please arrange the post-it notes in the order of frequency of use on your mobile device or tablet.                               |
| 8   | Please arrange the post-it notes according to the average percentage of revisited scraps.                                        |

### A.2 The number of people who used each scrap tool

**Table 3: The number of people who used each scrap tool**

| Location    | Tool                         | Respondents | Location    | Tool                     | Respondents |
|-------------|------------------------------|-------------|-------------|--------------------------|-------------|
| In-Browser  | Chrome – Bookmark            | 9           | In-Browser  | Safari – Bookmark        | 1           |
| In-Browser  | Safari – Favorites           | 4           | In-Browser  | Whale – Bookmark         | 1           |
| In-Browser  | Safari – Reading List        | 3           | In-Browser  | Whale – Scrapbook        | 1           |
| In-Browser  | Chrome – Reading List        | 2           | In-Browser  | Samsung - Bookmark       | 1           |
| Out-Browser | Kakaotalk - Chat with Myself | 10          | Out-Browser | PC Memo Application(iOS) | 2           |
| Out-Browser | Notion                       | 10          | Out-Browser | Naver Keep               | 2           |
| Out-Browser | Mobile Memo Application      | 6           | Out-Browser | Goodnote                 | 1           |
| Out-Browser | PC Sticky Notes              | 5           | Out-Browser | LiquidText               | 1           |
| Out-Browser | Onenote                      | 1           | Out-Browser | Google Drive             | 1           |
| Tool-Free   | Screenshot                   | 7           | Tool-Free   | Add to home in mobile    | 2           |
| Tool-Free   | Keep tabs opened             | 3           | Tool-Free   | Download as html         | 2           |

### A.3 Mental Model of Web Page Scraping Behavior

**Table 4: Complete version of mental model of web page scraping behavior**

| Mental Space | Description                | Task Tower                                                                                                          | Task<br>– <i>Atomic Task</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------|----------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Encounter    | User finds a new web page. | 1. Personal recommendation<br><br>2. Web surfing<br><br>3. SNS Following<br><br>4. Work(Study)                      | 1.1 In conversation<br>1.2 In lectures<br>1.3 In online chatting<br>2.1 Search for Shopping<br>2.2 Search for Hobby/Leisure<br>3.1 Online Community<br>3.2 Subscribe to paid contents<br>4.1 Research for work<br>4.2 Search for arising questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Create       | User Creates a Scrap       | 1. Determine whether to scrap the web page or not<br><br><br><br><br><br><br><br><br><br>2. Determine where to keep | 1.1 Read the page content<br>– 1.1.1 <i>Decide if the situation is suitable for reading</i><br>– 1.1.2 <i>Read title only</i><br>– 1.1.3 <i>Skimming</i><br>– 1.1.4 <i>Read all</i><br>1.2 Evaluate its perceived usefulness<br>– 1.2.1 <i>Potential future use</i><br>– 1.2.2 <i>Personal Interest</i><br>– 1.2.3 <i>General usefulness</i><br>1.3 Evaluate its importance<br>– 1.3.1 <i>Estimated storage duration</i><br>– 1.3.2 <i>Certainty for return</i><br>– 1.3.3 <i>Anticipated frequency of use</i><br>2.1 Available time remaining for creating<br>– 2.1.1 <i>Leave a memo while having conversation</i><br>2.2 Ease of use<br>– 2.2.1 <i>Prefers familiar tools</i><br>– 2.2.2 <i>Need of additional information</i><br>– 2.2.3 <i>Shareability for other devices</i><br>2.3 Future need of sharing with other people<br>– 2.3.1 <i>Need of publish</i><br>– 2.3.2 <i>Need of collaboration</i><br>2.4 Context of the web page<br>– 2.4.1 <i>Consider not only the page itself, but also the platform to which it belongs</i><br>– 2.4.2 <i>Consider a device suitable for the web page</i><br>– 2.4.3 <i>Consider the relationship with other scraps</i> |
| Revisit      | Revisit a Scrap            | 1. Intention to find a scrap arises<br><br><br><br>2. Search for the desired scrap                                  | 1.1 At the start and end of the day<br>1.2 New task starts in offline<br>1.3 Share information in communications<br>1.4 Needed for work<br>2.1 Keyword search<br>2.2 Locational navigation<br>– 2.2.1 <i>Read thumbnail, title, and paragraph preview</i><br>– 2.2.2 <i>Scroll down and check one-by-one</i><br>2.3 My own organizational hierarchy<br>2.4 Guess the purpose of a scrap with date<br>2.5 Lost in scraps while surfing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| Mental Space | Description            | Task Tower                      | Task – Atomic Task                                                                                                                                                                                                                                                                                |
|--------------|------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reuse        | Reuse the page content | 3. Explore a tool aimlessly     | 3.1 Aimlessly open a scrap tool<br>- 3.1.1 <i>At the start and end of the day</i><br>- 3.1.2 <i>On the road</i><br>- 3.1.3 <i>When get bored</i><br>- 3.1.4 <i>While working(studying)</i><br>3.2 Evaluate the importance of a scrap again<br>3.3 Rediscover scraps that didn't even know existed |
|              |                        | 1. Summarize/Organize           | 1.1 Add contextual and additional memo<br>1.2 Beautify relevant arrangements<br>1.3 Archive the URL after the arrangement.                                                                                                                                                                        |
|              |                        | 2. Share the summarization      | 2.1 Unspecified disclosure for viewing<br>2.2 Share directly with specific people                                                                                                                                                                                                                 |
|              |                        | 3. Get affective influence      | 3.1 Proud, satisfied<br>3.2 Indebtedness, shame, burden                                                                                                                                                                                                                                           |
| Organize     | Organize scrap lists   | 1. Transfer to an adequate tool | 1.1 Move from temporary tool to permanent tool<br>1.2 After moving, discard the scrap from temporary tool                                                                                                                                                                                         |
|              |                        | 2. Organize for findability     | 2.1 When creating, classify according to its theme<br>2.2 Sort hierarchically by topic<br>- 2.2.1 <i>Folding</i><br>- 2.2.2 <i>Toggle</i><br>- 2.2.3 <i>Database list</i><br>2.3. Change folder structure when interests change significantly                                                     |
|              |                        | 3. Get affective influence      | 3.1 Proud, satisfied<br>3.2 Indebtedness, shame, burden                                                                                                                                                                                                                                           |
|              |                        | 4. Delete a scrap               | 4.1. Discard non-critical scraps<br>4.2. Leave them till they no longer irritate eyes                                                                                                                                                                                                             |

## A.4 Data Source

Table 5: List of 10 media blogs for obtaining sample scrap data

| Number of Scraps | Category   | Website Domain                                                            |
|------------------|------------|---------------------------------------------------------------------------|
| 20               | Hobby      | <a href="https://abeautifulmess.com/">https://abeautifulmess.com/</a>     |
| 20               | Hobby      | <a href="https://nomadicmatt.com/">https://nomadicmatt.com/</a>           |
| 20               | Style      | <a href="https://theeverygirl.com/">https://theeverygirl.com/</a>         |
| 20               | Style      | <a href="https://mantelligence.com/">https://mantelligence.com/</a>       |
| 20               | News       | <a href="https://vice.com/">https://vice.com/</a>                         |
| 20               | News       | <a href="https://npr.com/">https://npr.com/</a>                           |
| 20               | Technology | <a href="https://discovermagazine.com/">https://discovermagazine.com/</a> |
| 20               | Technology | <a href="https://techcrunch.com/">https://techcrunch.com/</a>             |
| 20               | Trivia     | <a href="https://atlabobscura.com/">https://atlabobscura.com/</a>         |
| 20               | Trivia     | <a href="https://lifehacker.com/">https://lifehacker.com/</a>             |